Supplementary Material: Stage-Aware Early Warning of Australian Winter Crop Yield Shortfall

Code availability. Code and reproducibility scripts are available at:
`https://anonymous.4open.science/r/acml2026-crop-warning-review-E32F`

1. Fixed-Model and Feature-Regime Details

Table S1: Fixed-model lead-time comparison. NYH denotes the no-yield-history weather-soil regime; Op. denotes the operational regime. Values are test RMSE.

Window	Ridge NYH	Ridge Op.	LGBM NYH	LGBM Op.
May-Jun	1.01	0.74	0.92	0.86
May-Jul	1.00	0.72	1.06	0.85
May-Aug	1.01	0.75	0.98	0.89
May-Sep	0.96	0.70	0.94	0.89
May-Oct	0.89	0.66	0.91	0.83

Table S2: Detailed validation-selected lead-time performance by forecast window for the no-yield-history weather-soil and operational regimes. This table provides the numeric rows summarized by the main-paper lead-time RMSE comparison figure.

Window	Feature set	Model	MAE	RMSE	R^2
May-Jun	NYH weather-soil	XGBoost	0.651	0.914	0.478
May-Jul	NYH weather-soil	CatBoost	0.659	0.951	0.436
May-Aug	NYH weather-soil	HistGradientBoosting	0.609	0.968	0.416
May-Sep	NYH weather-soil	SVR-RBF	0.609	0.881	0.516
May-Oct	NYH weather-soil	SVR-RBF	0.548	0.821	0.579
May-Jun	Operational	ElasticNet	0.533	0.719	0.677
May-Jul	Operational	Ridge	0.527	0.716	0.680
May-Aug	Operational	ElasticNet	0.506	0.724	0.673
May-Sep	Operational	Ridge	0.469	0.704	0.691
May-Oct	Operational	Ridge	0.428	0.660	0.728

Table S3: Feature groups used for stage-aware early-warning models.

Feature group	Examples	Interpretation
Rainfall/dryness	rain_sum, rain_days, dry_days, dry-spell length	Water supply and dry-spell stress
Heat/cold	tmax_mean, heat_days_30, heat degree days, frost days	Heat exposure and frost/cold risk
Energy/evaporation	radiation_sum, evap_sum, vapor pressure	Growth energy and evaporative demand
Compound stress	hot-dry days, high-evaporation dry days	Combined weather stress
Soil background	AWC, clay/sand, SOC, pH, bulk density, depth	Regional vulnerability conditioning

Table S4: Feature-regime definitions used for ablation and information-source analysis.

Feature set	Weather	Soil	Yield hist.	Interpretation
Identity/time only	No	No	No	Crop-region persistence benchmark
Weather	Yes	No	No	Stage-weather signal only
Weather+Dev	Yes	No	No	Adds train-derived anomaly features
Weather+Soil	Yes	Yes	No	Adds static regional vulnerability
No-yield-history weather-soil	Yes	Yes	No	Main information-isolation regime
Yield history	No	No	Yes	Persistent productivity memory
Operational with yield history	Yes	Yes	Yes	Main deployment-oriented regime

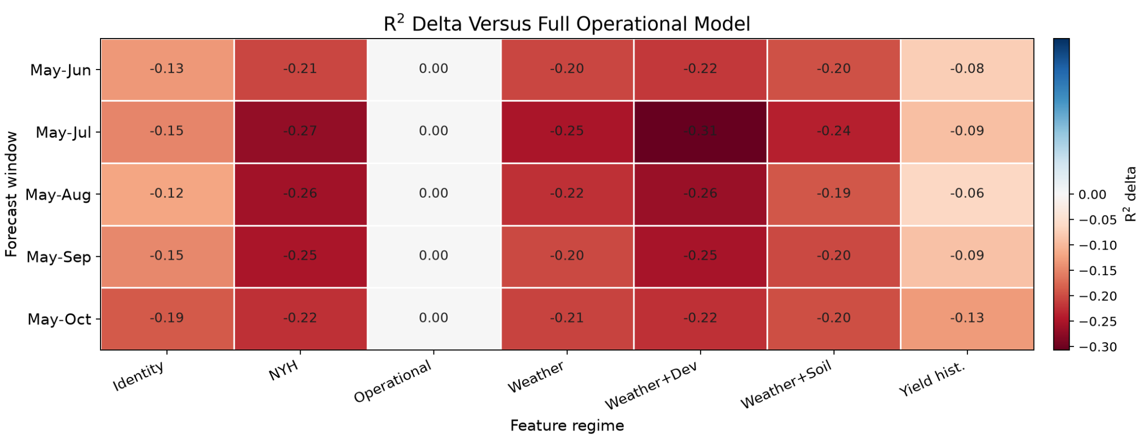

Figure S1: Ablation heatmap showing R^2 differences relative to the full operational feature set. Short labels are used to keep the dense comparison readable.

2. Data Coverage and Baseline Suite

Table S5: Missing source yield entries by state-crop series. The total confirms 966 observed entries out of 990 possible state-crop-year combinations in 1989–2021 for the observed crop-region pairs.

Region	Crop	Observed years	Missing years
New South Wales	Barley	33	0
New South Wales	Canola	33	0
New South Wales	Lupins	33	0
New South Wales	Oats	33	0
New South Wales	Wheat	33	0
Queensland	Barley	33	0
Queensland	Canola	30	3
Queensland	Lupins	24	9
Queensland	Oats	33	0
Queensland	Wheat	33	0
South Australia	Barley	33	0
South Australia	Canola	33	0
South Australia	Lupins	33	0
South Australia	Oats	33	0
South Australia	Wheat	33	0
Tasmania	Barley	33	0
Tasmania	Canola	33	0
Tasmania	Lupins	21	12
Tasmania	Oats	33	0
Tasmania	Wheat	33	0
Victoria	Barley	33	0
Victoria	Canola	33	0
Victoria	Lupins	33	0
Victoria	Oats	33	0
Victoria	Wheat	33	0
Western Australia	Barley	33	0
Western Australia	Canola	33	0
Western Australia	Lupins	33	0
Western Australia	Oats	33	0
Western Australia	Wheat	33	0
Total	All crops	966	24

Table S6: Full May-Oct strong-baseline suite under the same leakage-safe split. External crop-yield studies are used as methodological context, not as numerical baselines, because their crops, spatial scales, inputs, lead times, and evaluation protocols differ.

Group	Regime	Model	Yield hist.	MAE	RMSE	R^2
Historical	Past-only yield	3-year rolling mean	Yes	0.606	0.779	0.622
Historical	Past-only yield	Previous-year yield	Yes	0.562	0.784	0.616
Historical	Past-only yield	Crop-region train trend	Yes	0.613	0.823	0.577
Historical	Past-only yield	Crop-region train mean	Yes	0.683	0.961	0.423
Classical ML	NYH weather-soil	SVR-RBF	No	0.548	0.821	0.579
Classical ML	NYH weather-soil	Ridge	No	0.536	0.889	0.507
Classical ML	NYH weather-soil	ElasticNet	No	0.557	0.906	0.487
Classical ML	NYH weather-soil	RandomForest	No	0.685	1.012	0.360
Classical ML	Operational	Ridge	Yes	0.428	0.660	0.728
Classical ML	Operational	ElasticNet	Yes	0.439	0.671	0.719
Classical ML	Operational	SVR-RBF	Yes	0.500	0.757	0.642
Classical ML	Operational	RandomForest	Yes	0.564	0.836	0.564
Strong tabular ML	NYH weather-soil	XGBoost	No	0.583	0.888	0.508
Strong tabular ML	NYH weather-soil	CatBoost	No	0.591	0.901	0.493
Strong tabular ML	NYH weather-soil	LightGBM	No	0.581	0.907	0.486
Strong tabular ML	NYH weather-soil	HistGradientBoosting	No	0.605	0.925	0.466
Strong tabular ML	Operational	CatBoost	Yes	0.505	0.778	0.622
Strong tabular ML	Operational	XGBoost	Yes	0.515	0.779	0.621
Strong tabular ML	Operational	HistGradientBoosting	Yes	0.521	0.812	0.588
Strong tabular ML	Operational	LightGBM	Yes	0.532	0.826	0.575
Interpretable ML	NYH weather-soil	GAM	No	0.675	1.045	0.318
Interpretable ML	Operational	GAM	Yes	0.646	1.159	0.162
Sequence	Daily sequence NYH	DailyWeather-GRU	No	0.674	1.020	0.350
Sequence	Daily sequence operational	DailyWeather-GRU	Yes	0.520	0.718	0.678

All-window baseline-suite metrics remain available in the accompanying CSV artifacts; the PDF reports the full May-Oct suite to keep the comparison readable.

Table S7: Detailed stress-validation summary by validation protocol and feature regime. Rows report mean performance across the folds and forecast windows belonging to each protocol; the held-out-test row is an all-window stress summary, not the May-Oct-only result in the main lead-time comparison.

Protocol	Feature set	Model	Mean RMSE	Mean R^2	Folds
Leave-one-crop-out	Operational with yield history	Ridge	0.731	0.520	5
Leave-one-crop-out	No-yield-history weather-soil	Ridge	1.071	0.067	5
Leave-one-region-out	Operational with yield history	Ridge	0.823	-0.356	6
Leave-one-region-out	No-yield-history weather-soil	LightGBM	0.974	-0.285	6
Rolling-origin	Operational with yield history	Ridge	0.516	0.698	3
Rolling-origin	No-yield-history weather-soil	LightGBM	0.578	0.631	3
Held-out test years, all-window mean	Operational with yield history	Ridge	0.716	0.679	1
Held-out test years, all-window mean	No-yield-history weather-soil	LightGBM	0.963	0.420	1

3. Data and Implementation Details

The raw yield source is the ABARES Australian Crop Report state-crop-data workbook `03_AustCropRpt20260303_StateCropData_v1.0.0.xlsx`, accessed on 4 July 2026; the modeled panel is the derived `yield_panel.csv`. Fields used are crop area, production, and yield in tonnes per hectare, filtered to the five winter crops and six state-level regions in the main paper. The harvest year is represented by `year_start`. Production and area are retained for audit but excluded from model features.

SILO/LongPaddock weather inputs are daily rainfall, maximum and minimum temperature, radiation, evaporation, and vapor pressure. They are aggregated from May through each forecast cutoff. The system uses state-level regional daily series as a monitoring exposure proxy; latitude and longitude metadata are retained, but no crop-area-weighted gridded exposure is claimed. Weather-deviation features use training years only: for feature value $x_{r,w,t}$, the deviation is $x_{r,w,t} - \bar{x}_{r,w}^{train}$ for the same region and forecast window. Soil/ASRIS attributes are static regional background covariates from the Soil and Landscape Grid of Australia, including available water capacity, texture, bulk density, soil organic carbon, nutrients, pH, cation exchange capacity, and depth variables. These features condition regional vulnerability and may partly proxy region identity.

The no-yield-history regime has 73 predictors before one-hot expansion: crop, state-level region, year, 30 stage-weather summaries, 18 train-derived weather-deviation features, and 22 soil background attributes after excluding coordinate metadata. The operational regime adds one-year lag, three-year rolling past mean, and expanding past mean, for 76 predictors before one-hot expansion. Scaling, imputation, encoders, baselines, thresholds, and conformal residuals are fitted without test years; one-hot encoders handle unknown categories in stress tests. Configured weather thresholds are rain day 1 mm, heavy rain 10/25 mm, heat 25/30/35 C, frost 0 C, cold 5 C, and high evaporation 5 mm.

4. Baseline-Suite Implementation

The internal baseline suite is run once with random seed 42. This single-seed design is a limitation; model selection is by validation RMSE within each forecast window and feature regime, and test metrics are reported once. Tabular preprocessing uses median imputation and standard scaling for numeric predictors, one-hot encoding for crop and region, and unknown-category handling in held-out stress tests. The suite uses scikit-learn 1.9.0, XGBoost 3.3.0, LightGBM 4.6.0, CatBoost 1.2.10, pyGAM 0.12.0, and PyTorch 2.12.1+cpu in the recorded run.

Candidate models include Ridge with alpha 1.0 or 3.0; ElasticNet with alpha 0.01/l1 ratio 0.2 or alpha 0.03/l1 ratio 0.4; RandomForest with 400 trees and either unrestricted depth/min leaf 3 or depth 8/min leaf 5; SVR-RBF with C 1 or 10 and epsilon 0.1; HistGradientBoosting; XGBoost; LightGBM; CatBoost; and a pyGAM additive model over transformed features. The GRU sequence comparator uses daily weather up to the cutoff window, static covariates concatenated after the sequence encoder, and for the operational regime adds lag/rolling/expanding yield-history features to the static block. Two GRU configurations are validation-selected: hidden size 16 or 32, dropout 0.15 or 0.20, Adam learning rate 0.001, weight decay 1e-4, up to 160 epochs with patience 16.

5. Classification and Uncertainty Details

Table S8: Sensitivity of the low-yield-risk label to crop-specific training shortfall percentiles. The 80th percentile is the main setting.

Percentile	Train rate (%)	Validation rate (%)	Test rate (%)
70	30.1	13.8	24.2
80	20.3	11.2	18.1
90	10.4	2.6	9.4

Table S9: Best threshold-tuned classification metrics by forecast window. Pred.+ is the fraction of test examples flagged positive by the selected threshold.

Window	Model	Strategy	Thr.	Prec.	Recall	F1	ROC-AUC	PR-AUC	Brier	Pred.+
May-Jun	CatBoost	best F1	0.14	0.21	0.89	0.33	0.51	0.19	0.149	0.79
May-Jul	LightGBM	best F1	0.30	0.27	0.70	0.39	0.63	0.25	0.144	0.47
May-Aug	LogisticRegression	precision ≥ 0.3	0.22	0.27	0.78	0.40	0.65	0.25	0.147	0.52
May-Sep	HistGradientBoosting	best F1	0.63	0.28	0.74	0.41	0.69	0.27	0.197	0.48
May-Oct	LogisticRegression	precision ≥ 0.3	0.27	0.47	0.56	0.51	0.80	0.39	0.135	0.21

Table S10: Low-yield-risk confusion matrix using validation-selected thresholds.

Window	Model	Thr.	TN	FP	FN	TP	Prec.	Recall	F1
May-Oct	LogisticRegression	0.27	105	17	12	15	0.469	0.556	0.508

Table S11: Held-out May-Oct ranked watch-list example. Rank scores are uncalibrated classifier scores used for ordering analyst-review lists and should not be interpreted as fully calibrated event probabilities. Observed indicates the held-out low-yield-risk label.

Rank	Region	Crop	Year	Rank score	Alert	Observed	Shortfall
1	VIC	Oats	2018	1.000	1	1	0.465
2	NSW	Lupins	2018	1.000	1	1	0.410
3	VIC	Lupins	2018	1.000	1	0	0.274
4	VIC	Canola	2018	1.000	1	0	0.012
5	NSW	Wheat	2018	0.273	1	1	0.979
6	QLD	Wheat	2019	0.273	1	1	0.957
7	QLD	Wheat	2018	0.273	1	1	0.880
8	QLD	Barley	2019	0.273	1	1	0.865
9	TAS	Oats	2019	0.273	1	1	0.730
10	QLD	Canola	2018	0.273	1	1	0.646

Coverage below the nominal 80% target, especially in later windows, is interpreted as temporal calibration difficulty under small validation samples. These intervals are diagnostics for analyst review rather than reliable calibrated intervals for automatic decisions.

Table S12: Compact conformal uncertainty diagnostics by forecast window. Coverage and interval width are evaluated on held-out test years after validation-period calibration.

Window	Interval model	Nominal	Coverage	Width	qhat	Brier
May-Jun	Conformal ElasticNet	0.800	0.799	1.565	0.782	0.149
May-Jul	Conformal ElasticNet	0.800	0.792	1.423	0.711	0.144
May-Aug	Conformal ElasticNet	0.800	0.711	1.259	0.629	0.147
May-Sep	Conformal ElasticNet	0.800	0.758	1.251	0.626	0.197
May-Oct	Conformal ElasticNet	0.800	0.752	1.125	0.563	0.135

6. Supplementary Figures

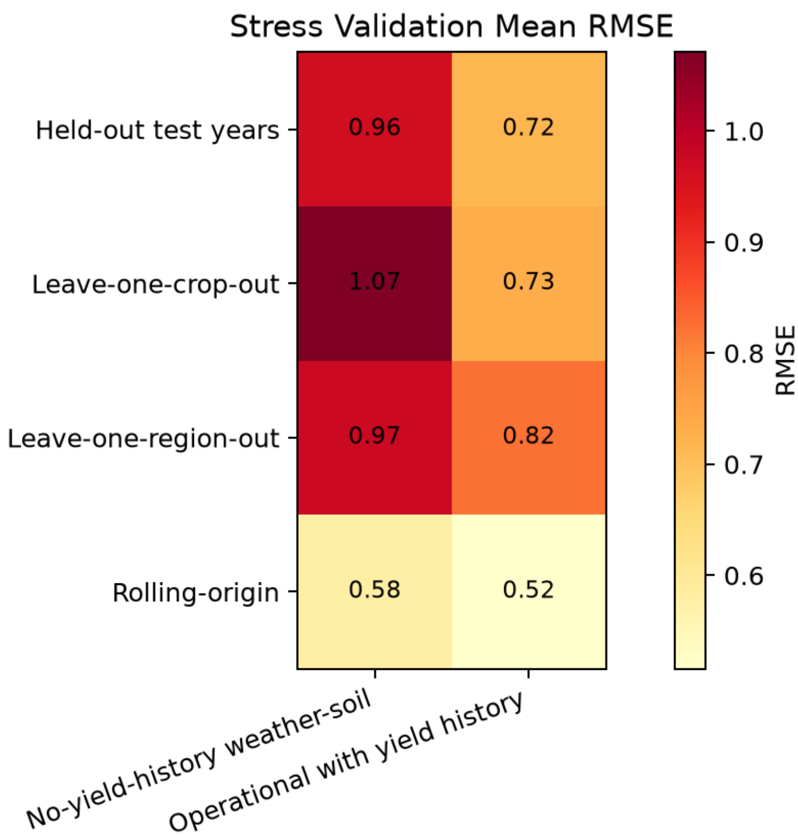

Figure S2: Stress validation heatmap by validation protocol and feature regime. Values use the same best-model, all-window summary as the main-paper stress validation table, not single-window May-Oct scores.

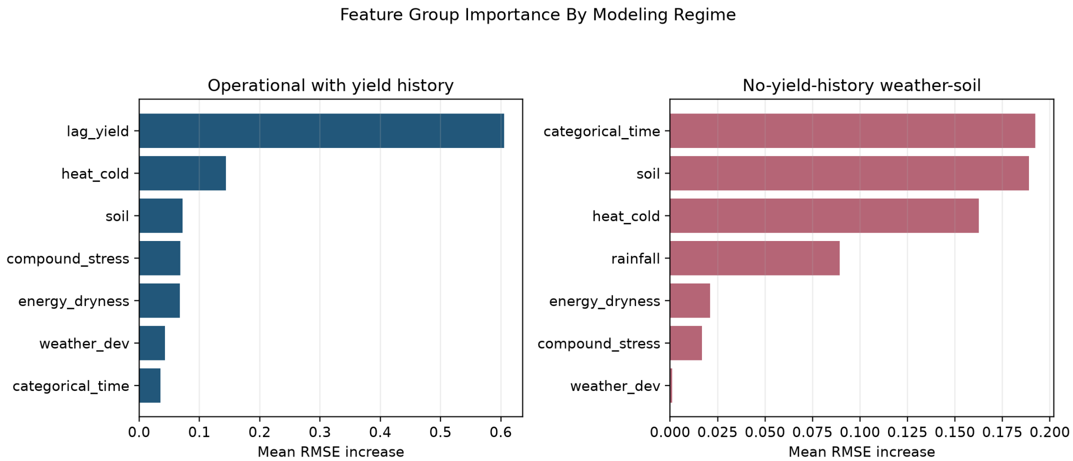

Figure S3: Feature group importance measured as mean RMSE increase after group permutation. Operational importance should not be interpreted as within-season weather importance because lag-yield features are allowed in that regime.